

Cavitoon Lite

World-space cavity for Unity URP

Cavitoon Lite is a full-screen World cavity effect for Unity 6 and the Universal Render Pipeline. It brightens convex details (Ridge) and darkens concave areas (Valley) while preserving the original camera alpha.

The effect uses camera depth and normals with a radius measured in scene units. It is suitable for stylized rendering, contact details, intersections, joints, modeling previews, and scenes that need stronger shape definition.

Edition Scope

Lite contains the complete World cavity algorithm with its full quality range. The output uses the classic monochrome Multiply response: Ridge brightens and Valley darkens the existing scene color.

Getting Started

Open **QuickStart.pdf** for installation steps and a recommended first preset. Cavitoon is installed as a URP Renderer Feature and does not require a custom material on every object.

Parameters

World Space

- **World Radius** - Search radius in Unity scene units. Match it to the scale of your objects.
- **World Ridge** - Strength of convex-surface brightening. Set to 0 to disable Ridge.
- **World Valley** - Strength of concave-surface darkening. Set to 0 to disable Valley.
- **World Attenuation** - Reduces the influence of samples farther from the center. Higher values localize the effect.
- **World Sample Count** - Quality level: 4, 8, or 16 radial depth samples per pixel.
- **Max World Pixel Radius** - Limits the projected radius near the camera. Set to 0 to disable the limit.

Rendering

- **Injection Point** - Selects where the pass runs. Before Rendering Transparents is the recommended default.
- **Show In Scene View** - Displays the effect in the Unity Scene view. Game Cameras are always supported.
- **Shader** - Must reference Hidden/Cavitoon/Cavity so the shader is retained in player builds.

Default Renderer Feature Values

Parameter	Default	Parameter	Default
World Radius	0.25	World Ridge	2.5
World Valley	1.0	World Attenuation	1.0
World Sample Count	16	Max Pixel Radius	24
Injection Point	Before Transparents	Scene View	Enabled

For a neutral starting point, set World Ridge and World Valley to 1.0. For faster rendering, begin with 8 samples and compare against 4 and 16 on the target hardware.

Rendering Requirements

- Unity 6 with the Universal Render Pipeline.
- URP Render Graph enabled. Legacy Compatibility Mode is not supported.
- An intermediate camera color texture plus valid camera depth and normals resources.
- Opaque materials with correct DepthOnly and DepthNormals passes.
- Standard URP Lit, Simple Lit, and Unlit materials are supported.
- Shader model 3.5 or newer on the target graphics API.

Cavitoon requests depth and normals through the Renderer Feature. Depending on the renderer and platform, URP may add a depth/normals prepass. The global URP Opaque Texture option is not required.

Performance

- Cost scales with render resolution, even when scene geometry is simple.
- World mode reconstructs 4, 8, or 16 depth samples for every affected screen pixel.
- Higher sample counts reduce sampling noise at an approximately proportional shader cost.
- Depth/normals generation and the camera-color copy can add bandwidth and extra passes.
- For mobile, start at 4 samples, keep the radius small, and profile on the target device.
- A lower Render Scale can reduce cost, but always verify the final visual result.

Limitations

- Cavitoon is screen-space: hidden, off-screen, and fully occluded geometry cannot contribute.
- The effect is designed for opaque geometry. Regular transparent materials do not generate cavity information.
- Custom shaders must keep alpha clipping, vertex deformation, and discard behavior consistent between their main, DepthOnly, and DepthNormals passes.
- Incorrect or missing normals may appear as unwanted bright or dark details.
- Non-uniform foveated rasterization is currently unsupported.
- Using an injection point after post-processing with an upscaler is not recommended when color and depth/normals use different resolutions.
- Preview and Reflection Cameras are skipped. Camera stacks should run Cavitoon on the camera that owns the opaque color, depth, and normals buffers.

Using Cavitoon with URP SSAO

World cavity and URP Screen Space Ambient Occlusion both darken recessed areas. Using strong values together may produce overly dark valleys. Temporarily disable SSAO while tuning Cavitoon, then reintroduce it at a lower intensity if needed.

Do not add a second Cavitoon Renderer Feature to increase strength. It doubles the full-screen cost and multiplies contrast. Adjust Ridge and Valley on a single feature instead.

Troubleshooting

No Effect Is Visible

- Confirm that the active camera uses the Renderer Data where Cavitoon was installed.
- Make sure the Cavitoon Renderer Feature is enabled.
- Disable URP Render Graph Compatibility Mode.
- Test with an opaque object using a standard URP Lit material.
- Confirm that the Shader field references Hidden/Cavitoon/Cavity.
- Increase World Ridge or World Valley temporarily to rule out an effect that is simply too weak.
- Match World Radius to the scale of the scene.
- Check the Console for missing color, depth, or normals resource warnings.

Black Spots or Excessively Dark Cavities

- Temporarily disable URP SSAO and compare the result.
- Reduce World Valley or World Radius, or increase World Attenuation.
- Check the camera near and far clip planes. An extreme depth range reduces precision.
- Verify that custom materials write correct depth and normals.

Bright Halos Around Silhouettes

- Reduce World Radius.
- Lower Max World Pixel Radius for objects that approach the camera.
- Check the material DepthNormals pass, especially with alpha clipping or vertex displacement.

Noise or Flickering

- Increase World Sample Count from 4 to 8 or 16.
- Reduce an excessively large radius or very high Ridge and Valley values.
- Compare with dynamic resolution and temporal anti-aliasing disabled to isolate the source.

Scene View Works but Game View Does Not

- Check the Renderer override on the Game Camera and the active quality level.
- Confirm that the quality tier uses the URP Pipeline Asset whose Renderer Data contains Cavitoon.
- For camera stacking, run the effect on the camera that produces the main opaque buffers.

UPGRADE TO CAVITOON PRO

Add Screen and Both modes, colored Ridge/Valley, Multiply/Additive/Overlay blending, Layer and Rendering Layer masks, plus Distance and Height Fade.

Get Cavitoon Pro: <https://assetstore.unity.com/packages/slug/399036>